SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 3 **Duration : 1 Hour**
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I M.Indu Priya(192372085)____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

   Constraints:

   i. D1 cannot work Night shift

   ii. D2 must work before D3 (Morning < Afternoon < Night)

   iii. Only one doctor per shift

   iv. D3 cannot work Morning

   v. All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

   Constraints:

   i. Movement allowed: Up, Down, Left, Right
   ii. Cost of each move = 1
   iii. Cannot pass through obstacles
   iv. Use Manhattan Distance heuristic

   Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

ANSWERS

1.Solution

Doctors: D1, D2, D3
Shifts: Morning (M), Afternoon (A), Night (N)

Constraints:

1. D1 ≠ Night

2. D2 must be before D3 (M < A < N)
3. Only one doctor per shift
4. D3 ≠ Morning
5. Each doctor gets exactly one shift

Step 1: Assign D1

Possible: M, A (not N)

Try **D1 = Morning**

Step 2: Assign D2

Remaining: A, N

Try **D2 = Afternoon**

Step 3: Assign D3

Remaining: N

Try **D3 = Night**

Constraint Check

- D1 ≠ Night → (Morning)
- D3 ≠ Morning → (Night)
- D2 before D3 → Afternoon < Night
- One doctor per shift
- All assigned

Final Valid Schedule

D1 → Morning
D2 → Afternoon
D3 → Night

2. Solution

Grid Path Finding using A* (Manhattan Distance Heuristic)

Given:

- Grid size: 5 × 5
- Start: S = (0,0)
- Goal: G = (4,4)
- Obstacles: (1,1), (1,2), (2,2), (3,1)
- Moves: Up, Down, Left, Right
- Cost per move = 1

Heuristic Function (Manhattan Distance)

$h(n) = |x_1 - x_2| + |y_1 - y_2|$

A* Evaluation Function

$f(n) = g(n) + h(n)$
$g(n)$ = cost from start
$h(n)$ = estimated cost to goal

Step-by-Step Path Selection (A*)

Start at **(0,0)**

1. (0,0) → f = 0 + 8 = 8
2. Move to (0,1) → f = 1 + 7 = 8
3. Move to (0,2) → f = 2 + 6 = 8
4. Move to (0,3) → f = 3 + 5 = 8
5. Move to (0,4) → f = 4 + 4 = 8
6. Move to (1,4) → f = 5 + 3 = 8
7. Move to (2,4) → f = 6 + 2 = 8
8. Move to (3,4) → f = 7 + 1 = 8
9. Move to (4,4) → f = 8 + 0 = 8

Final Path

(0,0) → (0,1) → (0,2) → (0,3) → (0,4) → (1,4) → (2,4) → (3,4) → (4,4)

Total Cost

= 8 moves

3. Solution

a) Analysis of Constraints

1. **Movement (Up, Down, Left, Right)**
→ The robot explores the grid step-by-step. No diagonal shortcuts, so path length may increase.
2. **Obstacles (Blocked Cells)**
→ The robot must avoid these cells. It may need to backtrack and find alternative paths.
3. **Limited Sensing (Adjacent Cells Only)**
→ The robot does not know the full map initially. It must explore and update knowledge while moving (partial observability).
4. **Movement Cost = 1**
→ Every step adds cost, so shorter paths are preferred.
5. **Risky Zones (Extra Cost = +2)**
→ Entering risky cells increases total cost (total = 3). The robot should avoid them unless necessary.
6. **Minimize Total Cost**
→ The robot must choose the safest and cheapest path, not just the shortest.
7. **Dynamic Knowledge (Online Search)**
→ The robot updates its map while moving and makes decisions in real-time.

b) Solution Approach (Online Uniform Cost Search)

Chosen Method: Online Search Agent using Uniform Cost Search (UCS)

Steps:

1. **Initialize**
    1. Start at position S
    2. Known map = empty
    3. Cost g(n) = 0
2. **Sense Environment**
    1. Observe adjacent cells (up, down, left, right)
    2. Mark cells as free, blocked, or risky
3. **Expand Possible Moves**
    1. Generate valid neighboring cells
    2. Ignore blocked cells
4. **Assign Cost**
    1. Normal cell = +1
    2. Risky cell = +3
5. **Select Next Move**
    1. Choose the path with **minimum cumulative cost (UCS principle)**
6. **Move and Update**
    1. Move to selected cell
    2. Update map with new observations
7. **Repeat**
    1. Continue until Goal (G) is reached

c) AI Concepts Applied

1. Intelligent Agent

- The robot acts as an **agent**:
    - **Percepts:** Adjacent cells (free, blocked, risky)
    - **Actions:** Move in 4 directions
    - **Goal:** Reach survivor safely

2. Rationality

The robot is **rational** because:

- It chooses actions that minimize total cost
- Avoids risky areas unless necessary
- Makes best decision based on available information

3. Environment Type

- **Partially Observable:** Cannot see entire grid
- **Deterministic:** Movement outcomes are predictable
- **Sequential:** Current action affects future steps
- **Dynamic:** Knowledge changes as robot explores
- **Discrete:** Grid-based environment

d) Justification of Approach

- **Uniform Cost Search (UCS)** ensures the **minimum-cost path**, considering both normal and risky cells.
- **Online Search** is necessary because:
  - The robot does not know the full environment initially
  - It must learn and adapt while moving
- This combination:
  - Handles **partial observability**
  - Minimizes cost effectively
  - Ensures safe navigation

Therefore, **Online UCS** is the most suitable approach for this problem.

Final Conclusion

The rescue robot uses an **online, cost-based search strategy** to dynamically explore the environment, avoid obstacles, minimize risk, and efficiently reach the survivor.